

WHEN TO STEER: RELIABILITY-ADAPTIVE PROBE GUIDANCE FOR CONTROLLABLE MUSIC DIFFUSION

Yushi Ye* Wilson Zheng* Yongyi Zang*

{yushiye, wilsonz}@andrew.cmu.edu, zyy0116@gmail.com

ABSTRACT

Latent-space probes provide lightweight gradient signals for steering diffusion-based music generation, but existing methods apply the resulting guidance on hand-picked subsets of sampling steps: recent work on Stable Audio Open intervenes either only in the first 20% of denoising or only in the last 60%, and neither choice is justified empirically. We ask *when* probe guidance should be applied. Recording real sampling trajectories of Stable Audio Open, we measure how pitch-class decodability of a 125k-parameter probe and its target-free confidence evolve over denoising time, and we measure causal steering sensitivity by intervening at single positions of the trajectory. The two peak at different stages: decodability keeps rising until denoising is nearly complete (88%), yet a guidance step changes the output most around the *middle* of sampling (54%), so where the probe is most accurate is not where it steers best. Building on this analysis we propose Reliability-Adaptive Probe Guidance (RAPG), which places a fixed budget of guidance updates at the steps of highest measured steering sensitivity and scales their strength with the probe’s target-free reliability. Across 15 prompts, 5 target melodies and 3 seeds, calibrating the schedule on steering sensitivity rather than decodability raises melodic coherence from 0.363 (the fixed window of prior work) to 0.445 at matched compute—a $3.5\times$ gain over unguided generation—while leaving prompt adherence (CLAP) and audio realism (Fréchet distance to real piano) unchanged. Most of this gain comes from the sensitivity-calibrated placement of the budget; reliability-based strength adaptation adds a smaller further improvement, and a reliability threshold alone, which follows decodability, adds none. *When* to steer is what the prior schedules got wrong.

Index Terms— controllable music generation, latent diffusion, training-free guidance, probing, guidance scheduling

1. INTRODUCTION

Text-to-music diffusion models such as Stable Audio Open (SAO) [1] produce high-quality audio but give the user only coarse control through the text prompt. Inference-time guidance adds finer control without retraining: a differentiable target loss is evaluated on the partially denoised sample and its gradient is used to nudge the latent [2, 3, 4, 5, 6]. For latent audio diffusion, back-propagating through the audio decoder is expensive, so two recent works replace the decoder-plus-feature-extractor by a small network that reads the control directly from the latent: latent-control heads (LatCH) [7] and pitch-class probes [8]. Both operate on SAO, both are cheap, and both steer the sampler with a fixed, manually chosen schedule.

Curiously, the two schedules are opposite. LatCH applies its “selective” guidance only during the first 20% of sampling steps,

arguing that later interventions drift off the data manifold [7]; the probe-guidance method of [8] waits until 40% of denoising is complete and then intervenes every second step, arguing that early latents are too noisy for the probe to be useful. Neither work ablates the choice. The question of *when* to steer is not specific to these two methods: limited-interval guidance is known to matter for classifier-free guidance in image diffusion [9], and training-free guidance frameworks expose the interval as a hyper-parameter [5], but for probe-based latent guidance of music there is no principled answer, and the answer is not obvious—the probe’s information content, the effectiveness of a gradient step and the damage it does to audio quality all change along the trajectory, and not necessarily in lockstep.

This paper studies that question on SAO with the pitch-class probe of [8], and turns the answer into a schedule that the sampler derives from its own latent state. Our prior work established that a lightweight latent probe *can* steer pitch-class content; here we ask *when* it should. Our contributions are:

1. **Trajectory-wise reliability analysis.** We record real SAO sampling trajectories and score the probe at every step against labels transcribed from the generated audio. Pitch-class decodability rises monotonically through denoising and reaches 95% of its maximum only at 88% of the trajectory, and the target-free confidence R_t of the probe tracks it closely (Pearson 0.96).
2. **Steerability $\neq$ decodability.** Decodability is an observational property of the latent; intervening at a single position of the trajectory measures instead where a guidance step actually changes the output. Steering sensitivity peaks near the middle of sampling (54%), far earlier than decodability saturates: a burst of updates at 54% changes melodic coherence several times more than the same burst applied late, even though the probe is much more accurate late. The early window used by prior selective guidance barely moves the output at all.
3. **RAPG.** A schedule that spends a fixed budget of K updates on the top- K steps of a calibrated profile (or, online, on steps passing a reliability threshold) and scales their strength by reliability. Calibrating on steering sensitivity rather than decodability improves the control/quality trade-off over every fixed window we tested; most of the gain is due to placement, strength adaptation adds a smaller improvement, and the online reliability threshold, which follows decodability, does not improve over the prior fixed window.

We also correct the target/latent alignment of the public implementation of [8] (Sec. 3); all numbers reported here use the corrected code. Code, schedules, per-run results and audio examples are available online.¹

*Equal contribution.

¹<https://stoneyey.github.io/mic-pitch-class-steering/>

2. METHOD

2.1. Probe-guided sampling

SAO [1] denoises a latent $\mathbf{z} \in \mathbb{R}^{64 \times T}$ ($T=1024$ frames at 21.5 Hz) with a diffusion transformer and an EDM-style sampler [10]; we use $N=50$ steps and classifier-free guidance [11] of 4.0. Step t denotes the latent returned by the t -th scheduler update, to which guidance is applied before the next denoising step, and we report positions as denoising progress $s=t/N \in [0, 1]$. A convolutional probe f_ϕ with 125k parameters, trained on VAE latents of MAESTRO [12] with Gaussian-noise augmentation, maps a latent to frame-level pitch-class probabilities $p_t = f_\phi(\mathbf{z}_t) \in [0, 1]^{12 \times T}$ [8]. Given a target pitch-class roll $\mathbf{y}^* \in \{0, 1\}^{12 \times T}$, the guidance loss is the binary cross-entropy $\mathcal{L}_t = \text{BCE}(p_t, \mathbf{y}^*)$ restricted to the audible frames, with gradient $\mathbf{g}_t = \nabla_{\mathbf{z}_t} \mathcal{L}_t$, and the latent is updated with the RMS-normalised, clamped step of [8]:

$$\mathbf{z}_t \leftarrow \mathbf{z}_t - \text{clamp}\left(\lambda_t \frac{\text{rms}(\mathbf{z}_t)}{\text{rms}(\mathbf{g}_t)} \mathbf{g}_t, \pm \alpha \text{rms}(\mathbf{z}_t)\right), \quad (1)$$

with $\alpha=0.05$. In [8], $\lambda_t = \lambda$ is constant and the update fires on a hand-chosen set of steps $\mathcal{T}$ (every second step after 40% of denoising, i.e. $K=15$ updates); in LatCH the set is the first 20% of steps. Everything that follows changes only $\mathcal{T}$ and λ_t .

2.2. Timestep reliability

We want a signal, available at every step without a target, that says how much the probe can be trusted on the current latent. The probe is multi-label (12 independent sigmoids), so the entropy of its raw outputs is not meaningful; worse, a probe that confidently predicts “no note anywhere” on pure noise would look reliable. We therefore normalise the outputs of each frame into a distribution $q_{t,f} = p_{t,f} / \sum_c p_{t,f,c}$ and measure how peaked it is:

$$R_t = 1 - \frac{1}{|\mathcal{F}|} \sum_{f \in \mathcal{F}} \frac{H(q_{t,f})}{\log 12} \in [0, 1], \quad (2)$$

where $\mathcal{F}$ is the set of audible frames. R_t is insensitive to the overall magnitude of p_t and costs one probe forward pass (<1 ms). Sec. 4.1 shows that R_t tracks the probe’s actual accuracy along real trajectories.

2.3. Reliability-adaptive probe guidance (RAPG)

RAPG answers *when* and *how strongly* to guide under a fixed budget of K updates. **How strongly:** the strength is scaled by reliability, $\lambda_t = \lambda R_t / \bar{R}$, where $\bar{R}$ is the mean reliability over guided steps on a held-out development set, so that the average strength equals the λ of the fixed baselines and only its distribution over time changes. **When (calibrated):** a per-step profile $w(t)$ is estimated once on the development set and the K steps with the largest $w(t)$ form $\mathcal{T}$. We consider two profiles—decodability $w(t) = F_1(t)$ and steerability $w(t) = \Delta C(t)$ (Sec. 4.2)—which lets us test directly whether the probe should be trusted where it is most accurate or where its gradient is most effective. **When (online):** guide at step t iff $R_t \geq \eta$ and fewer than K updates have been used, with η chosen on the development set so that the mean number of updates is K . The online rule adapts to each sample; the calibrated rule has zero overhead. Both keep the update rule of Eq. (1) unchanged.

3. EXPERIMENTAL SETUP

Base model and probe. SAO 1.0 in fp16, 50 sampling steps, CFG 4.0, 5-second outputs; the CNN probe checkpoint of [8] (micro- F_1 0.66 on held-out MAESTRO clips) is used unchanged. The SAO pipeline always denoises a 1024-frame latent (47.6 s) and crops the decoded audio to the requested length. We correct a temporal-scaling mismatch in the public implementation of [8], whose target construction spanned the full latent rather than the audible 5-s region (a 4-second melody was stretched over 38 s of latent time, so the audible seconds were steered towards its first note only): all methods here place the target at the true latent frame rate of 21.5 frames/s and evaluate the loss on the 108 audible frames, and “MLSP-fixed” denotes this corrected implementation with the original schedule.

Prompts, melodies, seeds. All reliability and sensitivity profiles, schedule choices and thresholds ($w(t)$, $\mathcal{T}$, η , $\bar{R}$) are calibrated exclusively on five development prompts; the evaluation prompts are held out and never used for schedule selection. The test set has 15 piano-centred prompts (the 9 of [8] plus 6 new), 5 eight-note target melodies at 120 BPM (4 s within each 5-s clip)—the ascending, descending and alternating patterns of [8] plus a repeated-note pedal and a leap-heavy zigzag—and 3 seeds, i.e. 225 generations per method, paired across methods.

Metrics. *Melodic coherence* is the pitch-class match rate of pYIN [13] frames inside target-active regions, exactly as in [8]; *chroma cosine* is the mean cosine between the CQT chroma of the output and the binary target; *CLAP* is the text-audio cosine of LAION-CLAP (clap-htsat-unfused) [14]; *FAD* is the Fréchet distance [15] between CLAP audio embeddings of each method’s outputs and of 600 MAESTRO clips. We report means with bootstrap 95% confidence intervals and paired Wilcoxon signed-rank tests (Holm-corrected).

Baselines. All guided methods use $K=15$ updates and $\lambda=0.05$: *Early* (steps 0–14, the LatCH-style early window), *Mid* (18–32), *Late* (35–49), *Uniform* (every ~ 3.5 steps), *MLSP-fixed* (20, 22, ..., 48), and RAPG in its calibrated ($F_1 / \Delta C$ profiles) and online variants; unguided SAO is the reference.

4. RESULTS

4.1. Reliability along real trajectories

Fig. 1(a) plots, over 50 unguided trajectories, the probe’s micro- F_1 against pitch-class labels transcribed from the generated audio, together with the target-free reliability R_t of Eq. (2). Decodability is near zero in the noisy early latents and rises monotonically, reaching 95% of its maximum only at 88% of the trajectory. R_t , computed without any target, follows the same curve (Pearson 0.96 across steps; median 0.85 within a single trajectory), which is what makes the online variant of RAPG possible. The Gaussian-noise augmentation used to train the probe [8] reproduces this curve well when the noise level is matched to each step, so decodability itself is not where the real and simulated trajectories diverge—steerability is.

4.2. Steering sensitivity

Decodability is observational: it says what can be read from a latent, not what happens to the output when the latent is pushed. To locate where guidance is *effective*, we intervene: a short burst of three updates ($\lambda=0.10$) is applied at a single position of the trajectory and the change in the final audio is measured, paired against the same-seed

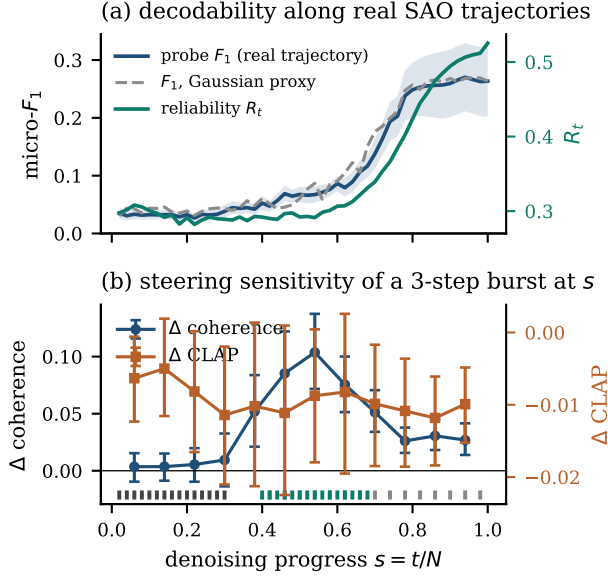


Fig. 1. (a) Probe reliability along real SAO trajectories: micro- F_1 against labels transcribed from the generated audio, the target-free reliability R_t of Eq. (2), and the Gaussian-noise proxy of [8] at matched noise level. (b) Causal steering sensitivity: change in melodic coherence and in CLAP score when guidance is applied only at one position of the trajectory (mean and 95% CI over 50 paired trials). Positions index the latent after the corresponding scheduler update. The strip marks the steps used by the Early, MLSP-fixed and RAPG schedules.

unguided generation (50 dev trials per position). Fig. 1(b) shows that the gain in melodic coherence peaks sharply near the middle of sampling (54%) and is several times smaller both early and late; a burst placed in the first 20% of steps—the interval used by selective guidance in [7]—changes coherence by less than a tenth of the mid-sampling effect. The peak is stable: over 5000 bootstrap resamples of the 50 trials the argmax lies at $s=0.54$ in 92% and within 0.46–0.62 in all of them, every leave-one-prompt-out and per-melody profile peaks in the same range, and the mid positions beat the early ones in 88% of individual trials (paired Wilcoxon $p < 10^{-7}$). Repeating the analysis at the $\lambda=0.05$ of the main experiments gives the same profile (peak at $s=0.54$, $\Delta C=0.099$ vs 0.104; Pearson 0.99 between the two curves; the top-15 steps shift by one, to 18–32), so the location of the peak does not depend on the burst strength. This is the opposite ordering from decodability, which is highest at the *end* of sampling. Consistently, the probe loss on the final latent is reduced most by *late* bursts, i.e. late guidance changes what the probe reads off the latent the most while changing the audio the least. A single burst leaves prompt adherence essentially unchanged at every position (Fig. 1(b), right axis); the quality question is settled at the full budget in Sec. 4.3.

4.3. Guidance window ablation and RAPG

Table 1 reports every schedule at the same budget of $K=15$ updates. Three findings stand out. First, placement is the dominant factor: at identical budget the early window barely improves over unguided SAO (0.17 vs 0.13), the mid window more than triples it

Table 1. Test set (15 prompts $\times$ 5 melodies $\times$ 3 seeds), all guided methods with $K=15$ updates ($n=225$ per row); schedules are calibrated on the disjoint development set. Means; bootstrap 95% CI half-widths are ≤ 0.035 for coherence and ≤ 0.017 for CLAP. $\dagger$: coherence significantly different from MLSP-fixed ($p < 0.05$, paired Wilcoxon, Holm). FAD is CLAP-embedding Fréchet distance to real piano; #u. is the number of probe updates.

Method	Coh. $\uparrow$	Chroma $\uparrow$	CLAP $\uparrow$	FAD $\downarrow$	#u.
SAO (unguided)	0.126 $\dagger$	0.22	0.307	0.82	0
Early (0–14)	0.166 $\dagger$	0.28	0.312	0.81	15
Mid (18–32)	0.424 $\dagger$	0.54	0.317	0.77	15
Late (35–49)	0.315 $\dagger$	0.46	0.311	0.81	15
Uniform	0.283 $\dagger$	0.41	0.314	0.79	15
MLSP-fixed	0.363	0.48	0.316	0.79	15
RAPG-cal (F_1)	0.310 $\dagger$	0.46	0.310	0.81	15
RAPG-cal (ΔC)	0.445 $\dagger$	0.56	0.324	0.75	15
RAPG-online	0.351	0.45	0.314	0.77	15.0
Top- K (ΔC), const. λ	0.434 $\dagger$	0.55	0.321	0.76	15

(0.42), and the late and MLSP windows fall in between—the guidance budget is the same, only its placement changes. Second, the two calibrated RAPG variants make the decodability/steerability gap concrete. Because $F_1(t)$ increases monotonically, the top- K steps of RAPG-cal(F_1) are exactly the late window (35–49), so its comparison with Late isolates reliability-based strength adaptation, which changes nothing there (0.310 vs 0.315); RAPG-cal(ΔC), calibrated on steering sensitivity, places the same budget mid-sampling and attains the highest coherence and chroma agreement of any method (0.445, vs MLSP-fixed 0.363, paired Wilcoxon $p < 10^{-7}$). Third, the control gain comes at no measurable quality cost: text–audio alignment is preserved (CLAP 0.324 for RAPG-cal(ΔC) vs 0.316 for MLSP-fixed and 0.307 for unguided SAO; no guided method differs significantly from MLSP-fixed), and the Fréchet distance to real piano remains comparable across methods (0.75 vs 0.79). The coherence gain is also not explained by reduced voicing: the fraction of pYIN-voiced frames is 0.88 under RAPG-cal(ΔC), against 0.86 for MLSP-fixed and 0.84 for unguided SAO. For every guided method, coherence exceeds the score the same audio would obtain if the clip were collapsed onto the melody’s first pitch class (e.g. 0.445 vs 0.255 for RAPG-cal(ΔC), 0.363 vs 0.263 for MLSP-fixed), confirming that the corrected guidance follows the time-varying melody rather than a single pitch class.

The advantage of RAPG-cal(ΔC) over MLSP-fixed holds for four of the five target melodies, from 0.372 vs 0.294 on the descending scale to 0.725 vs 0.607 on the repeated-note pedal, the pattern whose long single-pitch spans are easiest to steer; the two are within noise on the alternating arpeggio (0.327 vs 0.346). The on-line variant, which thresholds each sample’s own reliability, does not improve over MLSP-fixed (0.351 vs 0.363, $p=0.53$): because R_t tracks decodability, the threshold admits mostly late steps, and the result is consistent with our central finding that reliability identifies when pitch content becomes *decodable*, not when interventions are most *effective*. Adaptive strength contributes a small part of the gain: at the same mid window, scaling λ_t by reliability (RAPG-cal(ΔC), 0.445) improves only marginally over a constant λ (Top- K (ΔC), 0.434), so about 85% of the improvement over MLSP-fixed is obtained by placement alone; the benefit is in *where* the budget is spent, not how it is weighted.

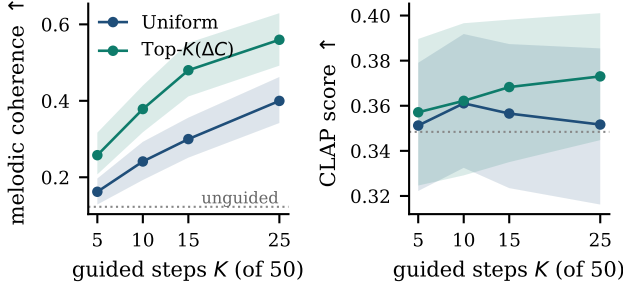


Fig. 2. Budget sweep on a paired 50-trial subset: melodic coherence (left) and CLAP score (right) against the number of guided steps for a uniformly spread budget and for the same budget placed at the steps of highest steering sensitivity (mean, bootstrap 95% CI). Dotted: unguided.

4.4. Budget sweep

Fig. 2 varies the number of guided steps, $K \in \{5, 10, 15, 25\}$, on a paired subset of the test set (10 prompts $\times$ 5 melodies, one seed) for the uniform schedule and for Top- $K(\Delta C)$ with constant λ , so that only the *placement* of the budget differs. The sensitivity-placed budget dominates at every K (paired Wilcoxon $p < 10^{-5}$ at each budget): ten sensitivity-selected updates recover 95% of the coherence of twenty-five uniformly spread ones (0.38 vs 0.40), and at $K=25$ coherence reaches 0.56 against 0.40. CLAP stays inside its confidence band for both schedules at every budget, so neither concentrating the updates nor adding more of them degrades prompt adherence in this range, and coherence has not saturated at $K=25$, which suggests the useful mid-sampling window is wider than the 15 steps used in Table 1. The sweep uses one seed per prompt–melody pair and is an efficiency analysis rather than the primary statistical comparison.

5. DISCUSSION AND CONCLUSION

The pitch content a probe can read from a latent and the pitch content a gradient step can *write* into the output are governed by different stages of denoising: readability grows until the latent is almost clean, but the mid-sampling steps, where coarse structure is still being decided, are where a nudge actually changes the music. A schedule tuned to the first signal (as decodability- or confidence-based heuristics implicitly are) spends its budget too late; tuning it to measured steering sensitivity recovers a markedly better control/quality trade-off at no extra compute, and recovers 95% of the coherence of 25 uniformly spread probe updates with 10 well-placed ones. Reliability is useful as a diagnostic and as a strength modulator, but not on its own as a placement criterion. The analysis is limited to one base model, to pitch-class control, and to guidance applied to $\mathbf{z}_t$ rather than the predicted clean latent $\hat{\mathbf{z}}_{0|t}$; whether the mid-sampling sweet spot generalises to other attributes and samplers, and perceptual validation of the CLAP and Fréchet-distance quality proxies, are left to future work. Still, that a target-free reliability signal predicts decodability along real trajectories and, as a strength modulator, adds to a sensitivity-placed budget suggests that trajectory-aware scheduling is a cheap, general lever for latent-space guidance.

6. COMPLIANCE WITH ETHICAL STANDARDS

This study used only publicly available pretrained models (Stable Audio Open) and datasets (MAESTRO) and generated synthetic audio; it involved no human or animal subjects, and no ethical approval was required.

7. REFERENCES

- [1] Zach Evans, Julian D Parker, CJ Carr, Zack Zukowski, Josiah Taylor, and Jordi Pons, “Stable audio open,” *arXiv preprint arXiv:2407.14358*, 2024.
- [2] Prafulla Dhariwal and Alexander Nichol, “Diffusion models beat GANs on image synthesis,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [3] Arpit Bansal, Hong-Min Chu, Avi Schwarzschild, Soumyadip Sengupta, Micah Goldblum, Jonas Geiping, and Tom Goldstein, “Universal guidance for diffusion models,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2023.
- [4] Jiwen Yu, Yinhuai Wang, Chen Zhao, Bernard Ghanem, and Jian Zhang, “FreeDoM: Training-free energy-guided conditional diffusion model,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- [5] Haotian Ye, Haowei Lin, Jiaqi Han, Minkai Xu, Sheng Liu, Yitao Liang, Jianzhu Ma, James Zou, and Stefano Ermon, “TFG: Unified training-free guidance for diffusion models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [6] Mark Levy, Bruno Di Giorgi, Floris Weers, Angelos Katharopoulos, and Tom Nickson, “Controllable music production with diffusion models and guidance gradients,” *Diffusion Models Workshop at NeurIPS*, 2023.
- [7] Zachary Novack, Zack Zukowski, CJ Carr, Julian Parker, Zach Evans, Josiah Taylor, Taylor Berg-Kirkpatrick, Julian McAuley, and Jordi Pons, “Low-resource guidance for controllable latent audio diffusion,” *arXiv preprint arXiv:2603.04366*, 2026.
- [8] Yushi Ye, Wilson Zheng, and Yongyi Zang, “Pitch-class steering for diffusion-based music generation via latent-space probes,” in *IEEE International Workshop on Machine Learning for Signal Processing (MLSP)*, 2026.
- [9] Tuomas Kynkäänniemi, Miika Aittala, Tero Karras, Samuli Laine, Timo Aila, and Jaakko Lehtinen, “Applying guidance in a limited interval improves sample and distribution quality in diffusion models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [10] Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine, “Elucidating the design space of diffusion-based generative models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [11] Jonathan Ho and Tim Salimans, “Classifier-free diffusion guidance,” *arXiv preprint arXiv:2207.12598*, 2022.
- [12] Curtis Hawthorne, Andriy Stasyuk, Adam Roberts, Ian Simon, Cheng-Zhi Anna Huang, Sander Dieleman, Erich Elsen, Jesse Engel, and Douglas Eck, “Enabling factorized piano music modeling and generation with the MAESTRO dataset,” in *International Conference on Learning Representations (ICLR)*, 2019.

- [13] Matthias Mauch and Simon Dixon, “pYIN: A fundamental frequency estimator using probabilistic threshold distributions,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2014, pp. 659–663.
- [14] Yusong Wu, Ke Chen, Tianyu Zhang, Yuchen Hui, Taylor Berg-Kirkpatrick, and Shlomo Dubnov, “Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.
- [15] Kevin Kilgour, Mauricio Zuluaga, Dominik Roblek, and Matthew Sharifi, “Fréchet audio distance: A reference-free metric for evaluating music enhancement algorithms,” in *Interspeech*, 2019.